

Bayesian modeling with brms

Outline

1. Comparison with lmer
2. Fixed effects
3. Fitting models in brms
4. Random effects
5. Reporting models

1. Comparison to Imer

Not Included

- Basic explanation of statistical concepts
- In-depth explanation of multilevel modeling concepts
- Model selection, troubleshooting, etc.

Refer to the Book!

- More thorough explanations for most things are in the book!
- Pointers to chapters.
- Happy to answer questions.

Setup (Ch. 1)

- The 'brms', 'bmmb', and 'lme4' R packages.

```
library (brms)
library (bmmb)
library (lme4)

options (contrast = c("contr.sum", "contr.sum"))
```

- The 'exp_data' dataset from the 'bmmb' R package, available in 'exp_data.csv'.

```
> head (exp_data)
  height  vtl C L S vtl_original
1  165.6 -1.2 g 1 1         12.2
2  165.6 -1.0 w 1 2         12.4
3  165.6 -1.8 g 1 3         11.6
4  154.9 -1.5 g 1 4         11.9
5  168.4 -1.3 b 1 5         12.1
6  147.8 -1.7 b 1 6         11.7
```

The Experiment (Ch. 1)

- Stimuli: the word 'heed' produced by 139 speakers from Hillenbrand et al. (1995).
- Speakers: Adult females, adult males, and boys and girls aged 10-12.
- Listeners: 15 native speakers of California English.
- Procedure: Listeners heard each syllable in random order and reported (i.e. guessed):
 - The height of the speaker in feet and inches (converted to cm).
 - The category of the speaker: boy, girl, man, woman.

Data (Ch. 1)

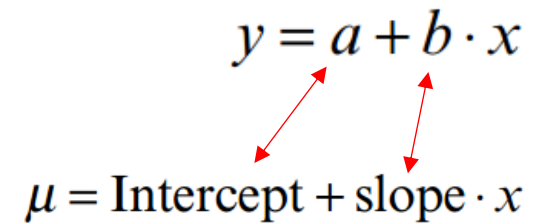
- The following variables:
 - **height**: apparent height in cm.
 - **vtl**: centered speaker vocal-tract length (in cm), estimated from speech acoustics.
 - **S**: a number indicating the speaker.
 - **L**: a number indicating the listener.
 - **C**: a letter indicating the apparent speaker category (b)oy, (g)irl, (m)an, or (w)oman.

Line Geometry (Ch. 9)

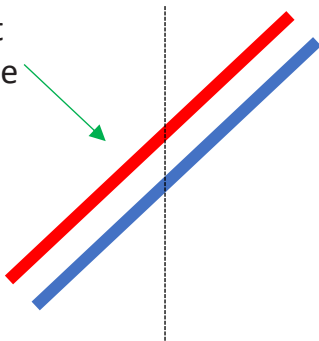
- Lines have two parameters:
 - An intercept determines the value of the line where $x = 0$.
 - A slope determines how much the line rises for every unit change in x .

$$y = a + b \cdot x$$

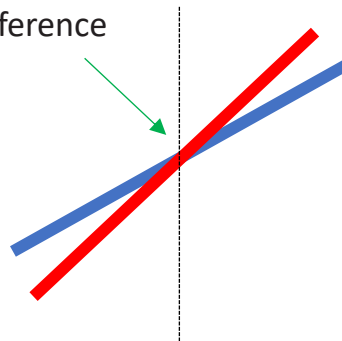
$\mu = \text{Intercept} + \text{slope} \cdot x$



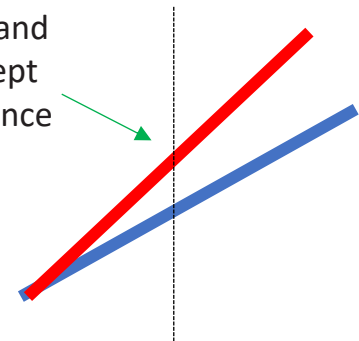
Intercept
difference



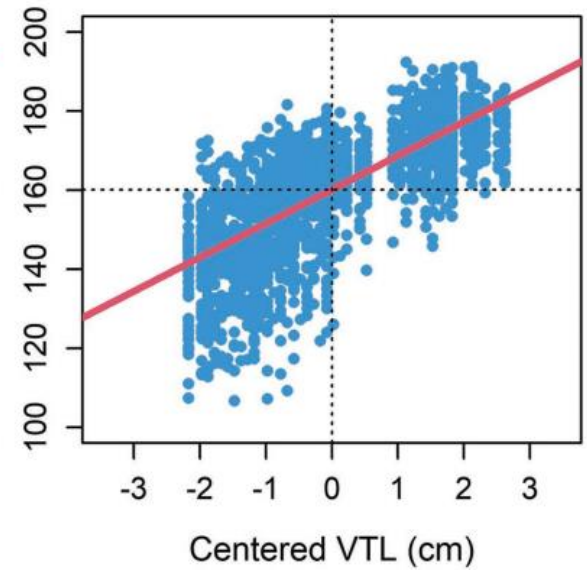
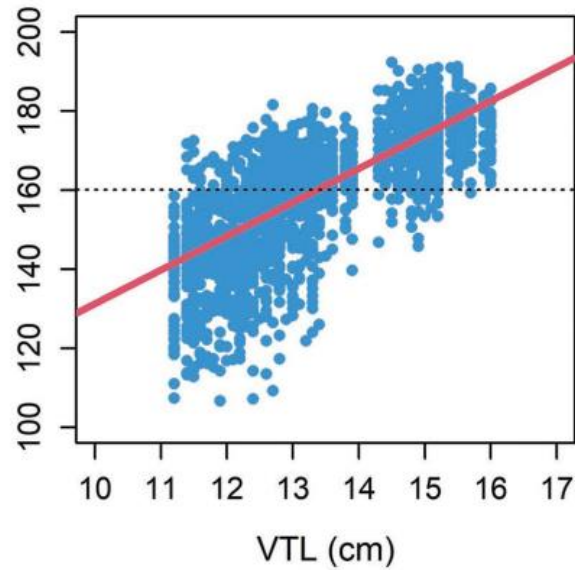
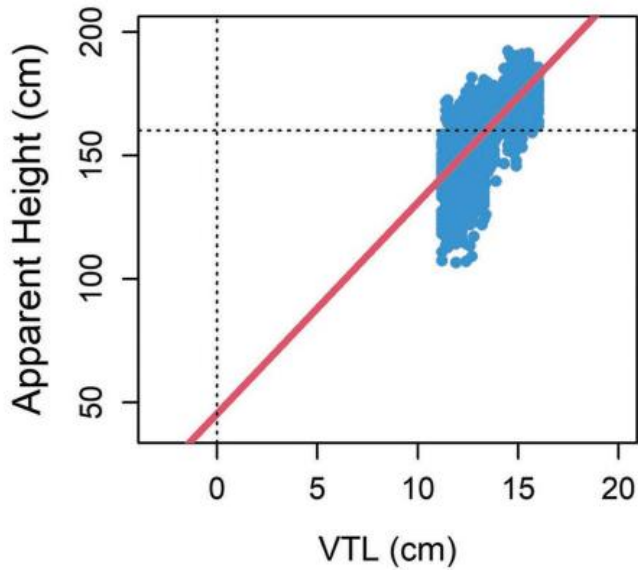
slope
difference



Slope and
Intercept
difference



Centering Predictors (Ch. 9)



Model (Ch. 9)

- Our model will have the following formula:

$$\text{height} \sim \text{vtl} * C + (\text{vtl} * C | L) + (1 | S)$$

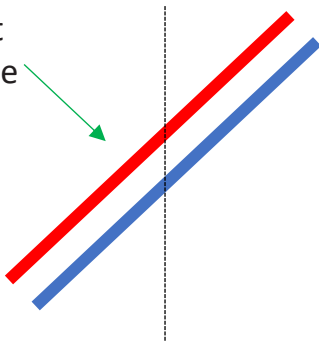
$$\text{height} \sim \text{vtl} + C + \text{vtl}:C + (\text{vtl} + C + \text{vtl}:C | L) + (1 | S)$$

- This includes:
 - vtl, a marginal vtl slope.
 - C, speaker group effects.
 - vtl:C, group-dependent vtl slopes
 - (vtl+C+vtl:C | L), listener intercept and random effects
 - (1 | S), a speaker random intercept

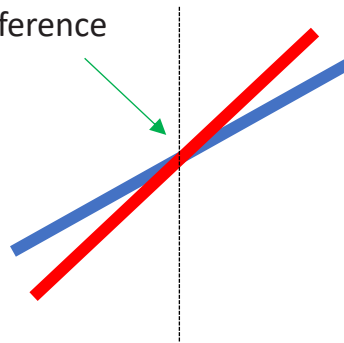
Line Geometry (Ch. 9)

- In our model:
 - Intercept and vtl are the marginal intercept and slope.
 - C reflects group-dependent changes in intercept.
 - vtl:C reflects group-dependent changes in vtl slope.

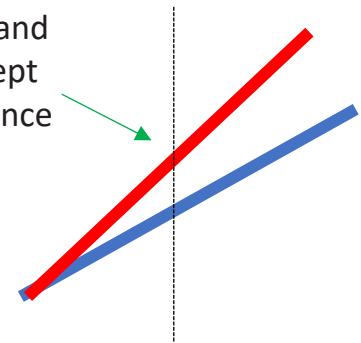
Intercept
difference



slope
difference



Slope and
Intercept
difference



lmer Model Fit

```
lmer_model = lme4::lmer (  
  height ~ vtl + C + vtl:C + (vtl + C + vtl:C|L) + (1|S),  
  data = exp_data)
```

boundary (singular) fit: see help('isSingular')

Random effects:

Groups	Name	Variance	Std.Dev.	Corr
S	(Intercept)	3.511	1.874	
L	(Intercept)	12.144	3.485	
	vtl	1.167	1.080	-0.03
	C1	27.551	5.249	0.72 -0.53
	C2	24.663	4.966	0.12 -0.14 -0.09
	C3	24.051	4.904	-0.38 0.59 -0.45 -0.80
	vtl:C1	6.438	2.537	0.12 0.28 0.32 -0.68 0.58
	vtl:C2	5.964	2.442	-0.02 0.84 -0.61 0.30 0.20 -0.24
	vtl:C3	3.527	1.878	-0.08 -0.91 0.31 0.43 -0.69 -0.41 -0.69
	Residual	47.088	6.862	

Number of obs: 2085, groups: S, 139; L, 15

Fixed effects:

	Estimate	Std. Error	t value
(Intercept)	160.1241	1.0141	157.899
vtl	4.1586	0.4391	9.470
C1	-8.5012	1.4653	-5.802
C2	-6.0501	1.5333	-3.946
C3	8.8682	1.5763	5.626
vtl:C1	0.8018	0.7961	1.007
vtl:C2	2.5826	0.8627	2.994
vtl:C3	-0.9315	0.7704	-1.209

lmer Model Fit

Random effects:

Groups	Name	Variance	Std.Dev.	Corr						
S	(Intercept)	3.511	1.874							
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	vt1	1.167	1.080	-0.03						
	C1	27.551	5.249	0.72	-0.53					
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	C3	24.051	4.904	-0.38	0.59	-0.45	-0.80			
	vt1:C1	6.438	2.537	0.12	0.28	0.32	-0.68	0.58		
	vt1:C2	5.964	2.442	-0.02	0.84	-0.61	0.30	0.20	-0.24	
	vt1:C3	3.527	1.878	-0.08	-0.91	0.31	0.43	-0.69	-0.41	-0.69

Residual 47.088 6.862

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?

Imer Model Structure

- Random effects are fit using some 'prior' information.
- It is assumed that:
$$S_i \sim N(0, \sigma_S)$$
- The effect of the random intercept for a speaker depends on:

$$P(y|S_i) \cdot P(S_i|\sigma_S)$$

Random effects:			
Groups	Name	Variance	Std.Dev.
S	(Intercept)	3.511	1.874
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	vtl	1.167	1.080
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vtl:C2	2.5826	0.8627	2.994
vtl:C3	-0.9315	0.7704	-1.209

Imer Model Structure

- When you have several for a 'cluster' you sample from a multivariate Gaussian distribution:

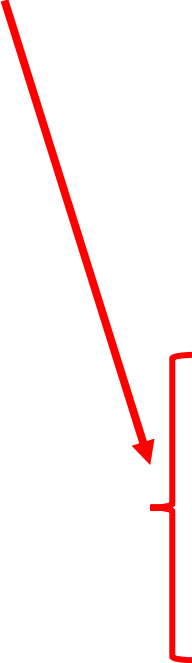
$$\vec{L}_i \sim MVN(0, \Sigma_S)$$

- The covariance matrix is broken up into correlations and standard deviations, not given priors.

Random effects:			
Groups	Name	Variance	Std.Dev.
S	(Intercept)	3.511	1.874
L	(Intercept)	12.144	3.485
	vt1	1.167	1.080
	C1	27.551	5.249
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lmer Model Structure

- There is no prior for any of the fixed effects.
- So, lmer uses a structure that is Bayesianesque for random effects, but sticks to maximum likelihood estimation (or something similar) for most things.



Random effects:			
Groups	Name	Variance	Std.Dev.
S	(Intercept)	3.511	1.874
L	(Intercept)	12.144	3.485
	vt1	1.167	1.080
	C1	27.551	5.249
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vt1:C2	2.5826	0.8627	2.994
vt1:C3	-0.9315	0.7704	-1.209

2. Fitting models in brms

brms Model Fitting (Ch. 3)

- Formula and data are specified the same as for lmer.
- Prior information can be specified for individual parameters, or for whole classes of parameters at a time.

```
brms_model = brm (  
  height ~ vt1 + C + vt1:C + (vt1 + C + vt1:C|L) + (1|S),  
  data = exp_data,  
  prior = c(set_prior("normal(162, 6)", class = "Intercept"),  
            set_prior("normal(0, 6)", class = "b"),  
            set_prior("normal(0, 6)", class = "sd"),  
            set_prior("lkj_corr(2)", class = "cor")),  
  chains = 4,  
  cores = 4,  
  thin = 1,  
  iter = 2000,  
  warmup = 1000  
)
```

brms Model Fitting (Ch. 3)

- Prior information can be specified for individual parameters, or for whole classes of parameters at a time.
- Classes are: Intercept, “b” (fixed effects), “sd” (random effect standard deviations), “cor” (random effect correlations).

```
brms_model = brm (  
  height ~ vtl + C + vtl:C + (vtl + C + vtl:C|L) + (1|S),  
  data = exp_data,  
  prior = c(set_prior("normal(162, 6)", class = "Intercept"),  
            set_prior("normal(0, 6)", class = "b"),  
            set_prior("normal(0, 6)", class = "sd"),  
            set_prior("lkj_corr(2)", class = "cor")),  
  chains = 4,  
  cores = 4,  
  thin = 1,  
  iter = 2000,  
  warmup = 1000  
)
```

brms Model Fitting (Ch. 3)

- Chains are individual series of samples. You can estimate these across multiple cores. Thinning keeps only every n^{th} sample.
- Warmup samples let your model calibrate and find good values to sample. The warmup is not kept by the model.

```
brms_model = brm (  
  height ~ vt1 + C + vt1:C + (vt1 + C + vt1:C|L) + (1|S),  
  data = exp_data,  
  prior = c(set_prior("normal(162, 6)", class = "Intercept"),  
            set_prior("normal(0, 6)", class = "b"),  
            set_prior("normal(0, 6)", class = "sd"),  
            set_prior("lkj_corr(2)", class = "cor")),  
  chains = 4,  
  cores = 4,  
  thin = 1,  
  iter = 2000,  
  warmup = 1000  
)
```

brms Model Output (Ch. 3)

- Basic information about model.
- L random effect sds and correlations.
- S random effect sd (no correlations)

```
> brms_model
Family: gaussian
Links: mu = identity; sigma = identity
Formula: height ~ vt1 + C + vt1:C + (vt1 + C + vt1:C | L) + (1 | S)
Data: exp_data (Number of observations: 2085)
Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
total post-warmup draws = 4000

Multilevel Hyperparameters:
~L (Number of levels: 15)

```

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
sd(Intercept)	3.65	0.74	2.47	5.31	1.00	1375	2232
sd(vt1)	1.23	0.47	0.40	2.31	1.00	1863	1130
sd(C1)	4.51	0.91	3.02	6.54	1.00	2243	2985
sd(C2)	3.85	0.98	2.24	6.11	1.00	2151	2835
sd(C3)	3.37	0.90	1.85	5.37	1.00	1563	2299
sd(vt1:C1)	2.39	0.72	1.22	4.01	1.00	2062	2074
sd(vt1:C2)	1.81	0.78	0.37	3.49	1.00	1367	1345
sd(vt1:C3)	0.73	0.53	0.03	1.95	1.00	1488	2409
cor(Intercept, vt1)	-0.18	0.24	-0.62	0.29	1.00	4431	3275
cor(Intercept, C1)	0.54	0.18	0.13	0.82	1.00	2626	2762
cor(vt1, C1)	-0.18	0.24	-0.62	0.32	1.00	2168	2695
cor(Intercept, C2)	0.12	0.21	-0.29	0.51	1.00	3772	2863
cor(vt1, C2)	-0.23	0.25	-0.68	0.27	1.00	1939	2390
cor(C1, C2)	-0.06	0.22	-0.47	0.37	1.00	3276	3339
cor(Intercept, C3)	-0.27	0.21	-0.65	0.16	1.00	3130	2622
cor(vt1, C3)	0.19	0.26	-0.32	0.67	1.00	1980	2583
cor(C1, C3)	-0.35	0.20	-0.69	0.09	1.00	3340	2714
cor(C2, C3)	-0.52	0.20	-0.83	-0.04	1.00	2842	3126
cor(Intercept, vt1:C1)	-0.04	0.22	-0.47	0.40	1.00	4393	3160
cor(vt1, vt1:C1)	0.19	0.25	-0.31	0.64	1.00	2352	2941
cor(C1, vt1:C1)	0.18	0.22	-0.27	0.59	1.00	3936	3390
cor(C2, vt1:C1)	-0.20	0.23	-0.62	0.28	1.00	3357	3215
cor(C3, vt1:C1)	0.25	0.23	-0.22	0.66	1.00	2494	3117
cor(Intercept, vt1:C2)	-0.03	0.24	-0.50	0.44	1.00	4744	3305
cor(vt1, vt1:C2)	0.22	0.27	-0.32	0.68	1.00	2950	2594
cor(C1, vt1:C2)	-0.16	0.25	-0.60	0.36	1.00	4936	3164
cor(C2, vt1:C2)	0.00	0.27	-0.54	0.51	1.00	3493	3014
cor(C3, vt1:C2)	0.11	0.25	-0.39	0.57	1.00	3499	3188
cor(vt1:C1, vt1:C2)	-0.14	0.27	-0.62	0.42	1.00	3513	3423
cor(Intercept, vt1:C3)	-0.06	0.29	-0.58	0.50	1.00	6144	2647
cor(vt1, vt1:C3)	-0.10	0.30	-0.63	0.49	1.00	4360	3243
cor(C1, vt1:C3)	-0.01	0.29	-0.58	0.54	1.00	4996	3219
cor(C2, vt1:C3)	0.11	0.30	-0.52	0.65	1.00	4612	3415
cor(C3, vt1:C3)	-0.12	0.30	-0.67	0.50	1.00	3775	3118
cor(vt1:C1, vt1:C3)	-0.07	0.29	-0.61	0.51	1.00	3988	3300
cor(vt1:C2, vt1:C3)	-0.13	0.30	-0.67	0.47	1.00	3019	3553

```
~S (Number of levels: 139)

```

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
sd(Intercept)	1.88	0.22	1.46	2.34	1.00	2277	3040

brms Model Output (Ch. 3)

- Information about fixed effects.
- The residual error.
- Boilerplate.

Regression Coefficients:

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
Intercept	160.53	1.05	158.51	162.65	1.00	1066	1962
vt1	4.16	0.49	3.19	5.14	1.00	2679	2968
C1	-8.25	1.26	-10.65	-5.71	1.01	1693	2546
C2	-5.35	1.30	-7.89	-2.75	1.00	2561	2552
C3	8.44	1.29	5.71	10.87	1.00	2087	2292
vt1:C1	0.49	0.80	-1.11	2.04	1.00	2874	2835
vt1:C2	2.88	0.78	1.32	4.43	1.00	3035	2859
vt1:C3	-1.07	0.67	-2.35	0.27	1.00	3355	2914

Further Distributional Parameters:

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
sigma	6.91	0.11	6.69	7.13	1.00	5940	3206

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS and Tail_ESS are effective sample size measures, and Rhat is the potential scale reduction factor on split chains (at convergence, Rhat = 1).

>

3. Fixed Effects

Inspect Fixed Effects

- Summary of posterior distribution of fixed effects.

```
> fixef (brms_model)
```

	Estimate	Est.Error	Q2.5	Q97.5
Intercept	160.5316656	1.0495805	158.505563	162.6526508
vt1	4.1620992	0.4898161	3.193425	5.1392794
c1	-8.2494423	1.2632459	-10.654192	-5.7051521
c2	-5.3536725	1.2979424	-7.890696	-2.7530721
c3	8.4407202	1.2894835	5.713325	10.8712773
vt1:c1	0.4884665	0.8027207	-1.111944	2.0351810
vt1:c2	2.8824961	0.7760545	1.315894	4.4286474
vt1:c3	-1.0651541	0.6653568	-2.350969	0.2714438

- Individual posterior samples.

```
> fixef_samples = fixef (brms_model, summary = FALSE)
> head (fixef_samples)
```

	variable							
draw	Intercept	vt1	c1	c2	c3	vt1:c1	vt1:c2	vt1:c3
1	160.6064	3.725294	-6.674870	-7.862968	8.940368	-0.04450138	2.711293	-0.6265771
2	161.6406	4.372845	-7.430997	-6.888431	7.717760	-0.55368148	4.003368	-1.6492226
3	160.6192	4.100990	-6.552364	-5.015939	6.696979	-0.10093706	1.718890	-0.5709548
4	163.3624	3.605621	-7.643150	-6.305637	9.895304	1.38994999	3.460993	-1.3436385
5	161.8735	4.446902	-6.735912	-5.613137	7.637109	0.90715703	2.491939	-1.0816397
6	161.7692	3.989583	-6.109933	-5.341421	8.361035	0.19806606	2.573578	-1.0367351

lmer vs. brms Fixed Effects

```
> fixef (brms_model)
```

	Estimate	Est.Error	Q2.5	Q97.5
Intercept	160.5316656	1.0495805	158.505563	162.6526508
vt1	4.1620992	0.4898161	3.193425	5.1392794
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vt1:C1	0.4884665	0.8027207	-1.111944	2.0351810
vt1:C2	2.8824961	0.7760545	1.315894	4.4286474
vt1:C3	-1.0651541	0.6653568	-2.350969	0.2714438

Fixed effects:

	Estimate	Std. Error	t value
(Intercept)	160.1241	1.0141	157.899
vt1	4.1586	0.4391	9.470
C1	-8.5012	1.4653	-5.802
C2	-6.0501	1.5333	-3.946
C3	8.8682	1.5763	5.626
vt1:C1	0.8018	0.7961	1.007
vt1:C2	2.5826	0.8627	2.994
vt1:C3	-0.9315	0.7704	-1.209

Fixed Effects (Ch. 9)

	Estimate	Std. Error	t value
Fixed effects:			
(Intercept)	160.1241	1.0141	157.899
vt1	4.1586	0.4391	9.470
C1	-8.5012	1.4653	-5.802
C2	-6.0501	1.5333	-3.946
C3	8.8682	1.5763	5.626
vt1:C1	0.8018	0.7961	1.007
vt1:C2	2.5826	0.8627	2.994
vt1:C3	-0.9315	0.7704	-1.209

Intercept
main effect

Slope main effect

Category-dependent
intercepts

Category-
dependent slopes

Combining Parameters (Ch. 5)

- The 'hypothesis' (or 'short_hypothesis') function can be used to combine parameters.

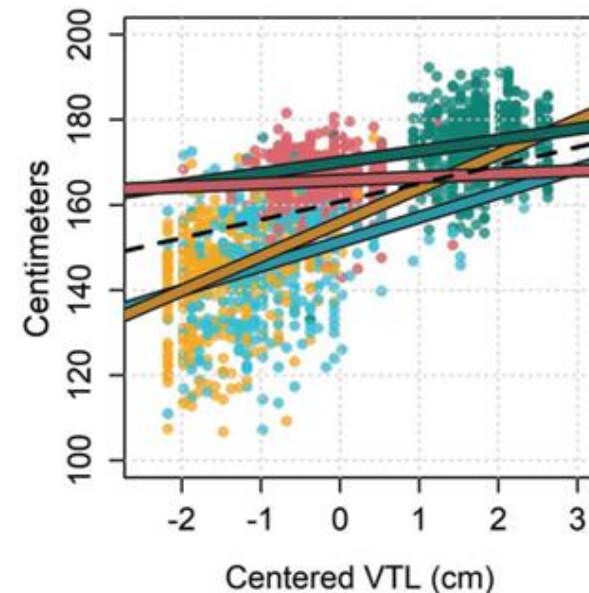
```
> hypothesis (brms_model, c("C1=0", "C2=0", "C3=0", "-(C1+C2+C3)=0"))
Hypothesis Tests for class b:
      Hypothesis Estimate Est.Error CI.Lower CI.Upper Evid.Ratio Post.Prob Star
1      (C1) = 0    -8.25      1.26   -10.65    -5.71         NA        NA    *
2      (C2) = 0    -5.35      1.30    -7.89    -2.75         NA        NA    *
3      (C3) = 0     8.44      1.29     5.71    10.87         NA        NA    *
4 -(C1+C2+C3) = 0     5.16      1.29     2.63     7.78         NA        NA    *
---
'CI': 90%-CI for one-sided and 95%-CI for two-sided hypotheses.
'*': For one-sided hypotheses, the posterior probability exceeds 95%;
for two-sided hypotheses, the value tested against lies outside the 95%-CI.
Posterior probabilities of point hypotheses assume equal prior probabilities.
```

```
> bmb::short_hypothesis (brms_model, c("C1=0", "C2=0", "C3=0", "-(C1+C2+C3)=0"))
      Estimate Est.Error      Q2.5      Q97.5      hypothesis
H1 -8.249442  1.263246 -10.654192 -5.705152      (C1) = 0
H2 -5.353673  1.297942  -7.890696 -2.753072      (C2) = 0
H3  8.440720  1.289483   5.713325 10.871277      (C3) = 0
H4  5.162395  1.293502   2.634388  7.776114 -(C1+C2+C3) = 0
```

Combining Parameters

- This can be used to find the group means and slopes from the model parameters.

```
> short_hypothesis (
+   brms_model,
+   c("Intercept = 0",
+     "Intercept + C1 = 0",
+     "Intercept + C2 = 0",
+     "Intercept + C3 = 0",
+     "Intercept + -(C1+C2+C3) = 0",
+     "vtl = 0",
+     "vtl + vtl:C1 = 0",
+     "vtl + vtl:C2 = 0",
+     "vtl + vtl:C3 = 0",
+     "vtl + -(vtl:C1+vtl:C2+vtl:C3) = 0"))
+   Estimate Est.Error      Q2.5      Q97.5 hypothesis
H1  160.531666  1.0495805  158.505563  162.652651 (Intercept) = 0
H2  152.282223  1.8929696  148.642639  155.974664 (Intercept+C1) = 0
H3  155.177993  1.7857377  151.649829  158.551611 (Intercept+C2) = 0
H4  168.972386  1.6507737  165.721152  172.255450 (Intercept+C3) = 0
H5  165.694060  1.2351481  163.197705  168.210223 (Intercept+-(C1+C2+C3)) = 0
H6   4.162099  0.4898161   3.193425   5.139279 (vtl) = 0
H7   4.650566  0.9327732   2.840533   6.435712 (vtl+vtl:C1) = 0
H8   7.044595  0.9560694   5.142744   8.940168 (vtl+vtl:C2) = 0
H9   3.096945  0.8273995   1.429100   4.734275 (vtl+vtl:C3) = 0
H10  1.856291  1.0804853  -0.301442   4.021505 (vtl+-(vtl:C1+vtl:C2+vtl:C3)) = 0
```



Combining Parameters (Ch. 5)

- You can also combine columns of samples ‘manually’.

```
> group_means = cbind(
+   fixef_samples[,"Intercept"]+fixef_samples[,"C1"],
+   fixef_samples[,"Intercept"]+fixef_samples[,"C2"],
+   fixef_samples[,"Intercept"]+fixef_samples[,"C3"],
+   fixef_samples[,"Intercept"]+ - (fixef_samples[,"C1"]+fixef_samples[,"C2"]+fixef_samples[,"C3"])
+ )
>
> head (group_means)
      [,1]      [,2]      [,3]      [,4]
1 153.9316 152.7435 169.5468 166.2039
2 154.2096 154.7522 169.3584 168.2423
3 154.0668 155.6032 167.3161 165.4905
4 155.7192 157.0567 173.2577 167.4158
5 155.1376 156.2604 169.5106 166.5855
6 155.6592 156.4277 170.1302 164.8595
>
> posterior_summary (group_means)
      Estimate Est.Error      Q2.5      Q97.5
[1,] 152.2822   1.892970 148.6426 155.9747
[2,] 155.1780   1.785738 151.6498 158.5516
[3,] 168.9724   1.650774 165.7212 172.2555
[4,] 165.6941   1.235148 163.1977 168.2102
```

Combining Parameters

- This yields identical results to using the hypothesis function.

```
> group_means = cbind(
+   fixef_samples["Intercept"]+fixef_samples
+   fixef_samples["Intercept"]+fixef_samples
+   fixef_samples["Intercept"]+fixef_samples
+   fixef_samples["Intercept"]+ - (fixef_sar
+ )
>
> head (group_means)
      [,1]      [,2]      [,3]      [,4]
1 153.9316 152.7435 169.5468 166.2039
2 154.2096 154.7522 169.3584 168.2423
3 154.0668 155.6032 167.3161 165.4905
4 155.7192 157.0567 173.2577 167.4158
5 155.1376 156.2604 169.5106 166.5855
6 155.6592 156.4277 170.1302 164.8595
>
> posterior_summary (group_means)
      Estimate Est.Error    Q2.5    Q97.5
[1,] 152.2822  1.892970 148.6426 155.9747
[2,] 155.1780  1.785738 151.6498 158.5516
[3,] 168.9724  1.650774 165.7212 172.2555
[4,] 165.6941  1.235148 163.1977 168.2102
```

```
> short_hypothesis (
+   brms_model,
+   c("Intercept = 0",
+     "Intercept + C1 = 0",
+     "Intercept + C2 = 0",
+     "Intercept + C3 = 0",
+     "Intercept + -(C1+C2+C3) = 0",
+     "vt1 = 0",
+     "vt1 + vt1:C1 = 0",
+     "vt1 + vt1:C2 = 0",
+     "vt1 + vt1:C3 = 0",
+     "vt1 + -(vt1:C1+vt1:C2+vt1:C3) = 0")) # gr
      Estimate Est.Error    Q2.5    Q97.5
H1 160.531666  1.0495805 158.505563 162.652651
H2 152.282223  1.8929696 148.642639 155.974664
H3 155.177993  1.7857377 151.649829 158.551611
H4 168.972386  1.6507737 165.721152 172.255450
H5 165.694060  1.2351481 163.197705 168.210223
H6  4.162099  0.4898161  3.193425  5.139279
H7  4.650566  0.9327732  2.840533  6.435712
H8  7.044595  0.9560694  5.142744  8.940168
H9  3.096945  0.8273995  1.429100  4.734275
H10 1.856291  1.0804853 -0.301442  4.021505 (v
```

Working with Posteriors

- You can find the probability that a parameter has a given values or inspect differences between parameters.

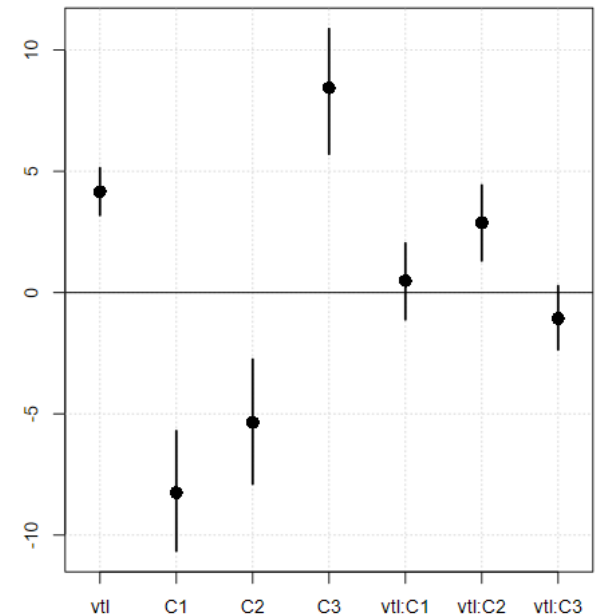
```
> mean (group_means[,1] < 150)
[1] 0.10875
> mean (group_means[,1] - group_means[,2])
[1] -2.89577
> mean ((group_means[,1] - group_means[,2]) > 0)
[1] 0.06975
> posterior_summary (group_means[,1] - group_means[,2])
      Estimate Est.Error      Q2.5      Q97.5
[1,] -2.89577  1.910027 -6.662023  0.8966352
```


brmplot

```
> fixef (brms_model)
```

	Estimate	Est.Error	Q2.5	Q97.5
Intercept	160.5316656	1.0495805	158.505563	162.6526508
vtl	4.1620992	0.4898161	3.193425	5.1392794
C1	-8.2494423	1.2632459	-10.654192	-5.7051521
C2	-5.3536725	1.2979424	-7.890696	-2.7530721
C3	8.4407202	1.2894835	5.713325	10.8712773
vtl:C1	0.4884665	0.8027207	-1.111944	2.0351810
vtl:C2	2.8824961	0.7760545	1.315894	4.4286474
vtl:C3	-1.0651541	0.6653568	-2.350969	0.2714438

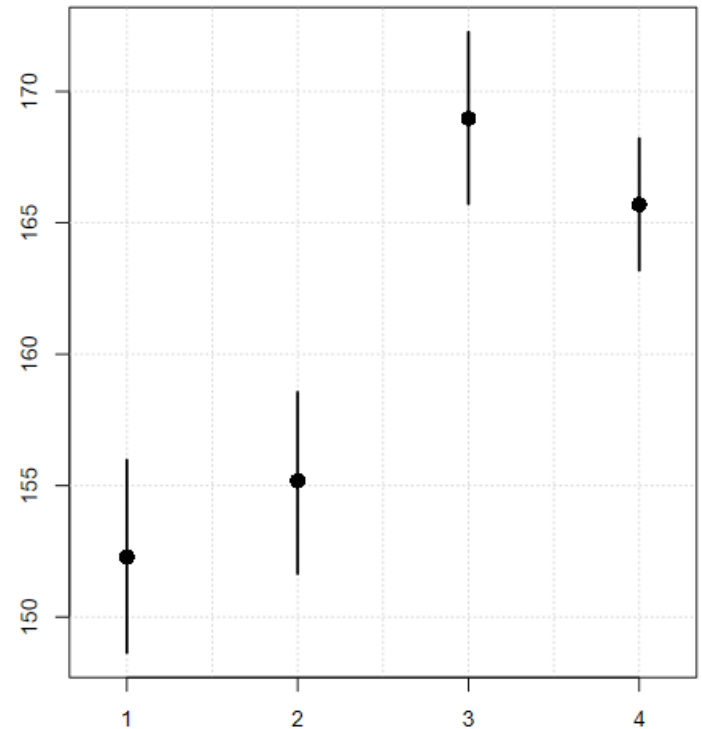
- Function takes in 4-column dataframes and makes plots of posterior distributions.



brmplot

```
mean_summary = posterior_summary (group_means)  
bmb::brmplot (mean_summary)
```

- You can make these dataframes with the `posterior_summary` function from `brms`.



lmer and brms comparison

```
pts = lme4::fixef (lmer_model)[-1]
errBars = summary (lmer_model)$coefficients[-1,2]

fixefs = brms::fixef (brms_model)[-1,]

par (mfrow = c(1,1), mar = c(5,4,1,1))
bmb::brmplot (brms::fixef (brms_model)[-1,], ylim = c(-13,13), las=2,
              pch=pchs,lwd=2, ylab = "Centimeters")
points (brms::fixef (brms_model)[-1,1], pch=16,lwd=2,cex=1.5)
points ((1:7)+.2, pts, cex=1.5,lwd=2,col=2,pch=16)
segments((1:7)+.2, pts-2*errBars,(1:7)+.2, pts+2*errBars,lwd=2,col=2)
```

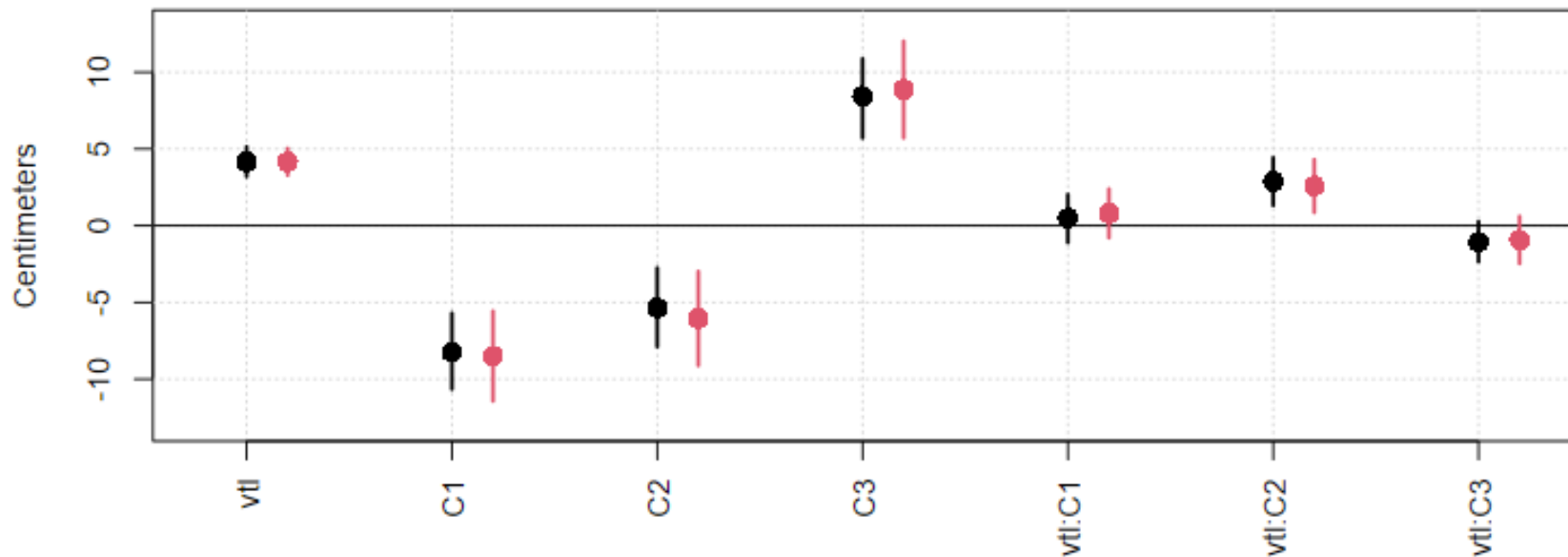
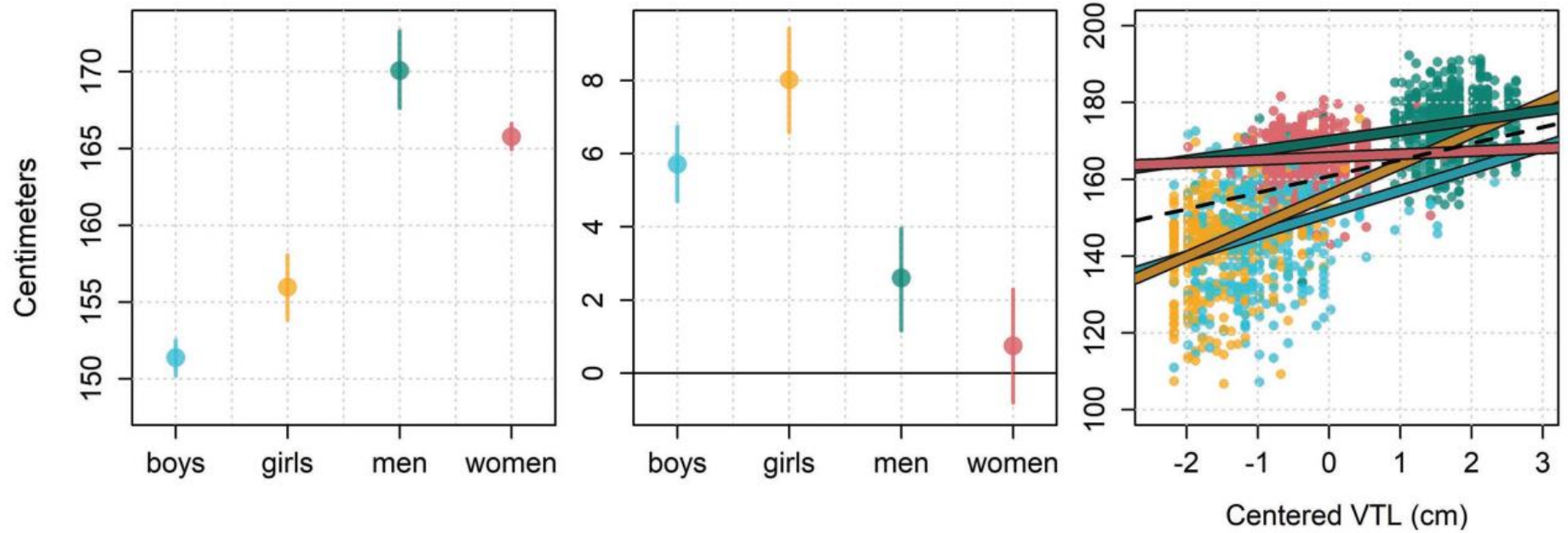


Figure 9.5, Chapter 9



Exercise

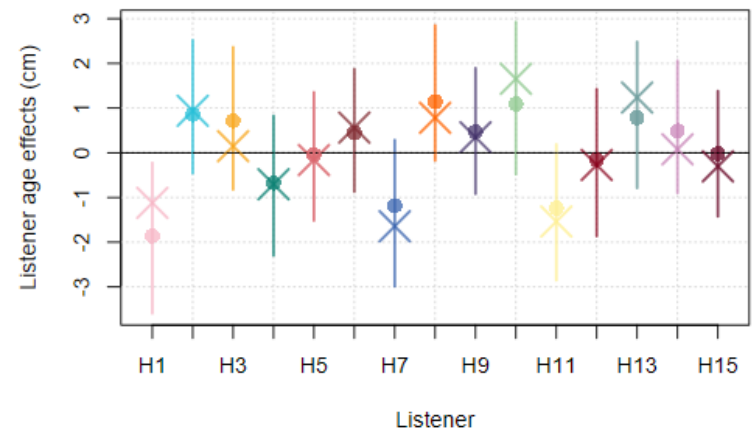
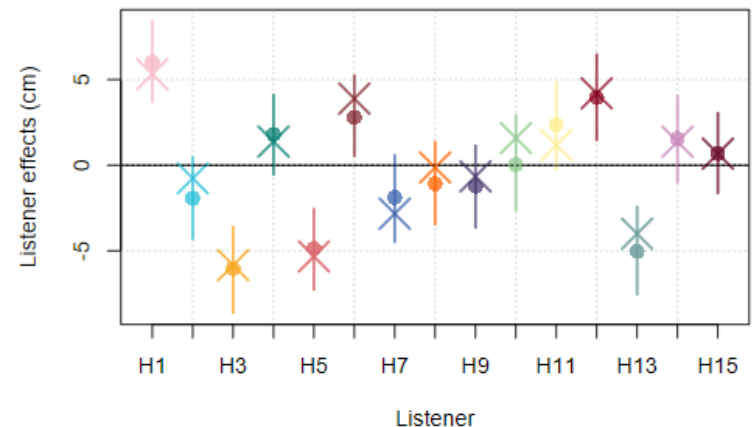
1. Find the differences between all pairs of groups.
2. Make a plot of these differences.
3. Which differences do you think are 'real'? Why?

4. Random Effects

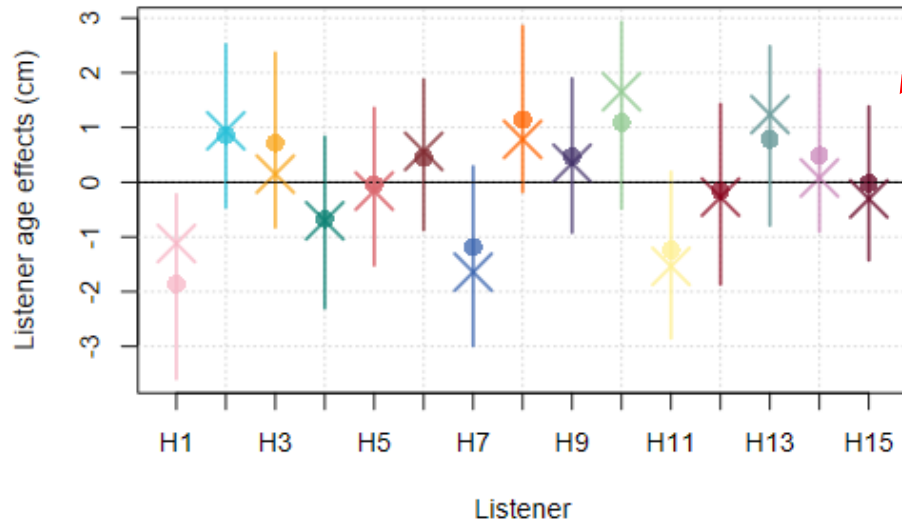
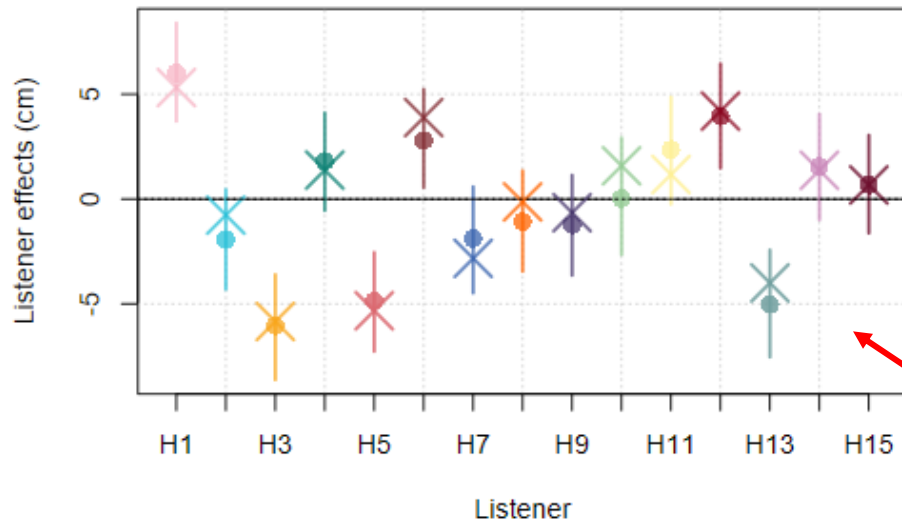
Random Effects (Ch. 5)

- You can get random effects with the hypothesis function.
- You get intervals for all parameters!!

```
random_intercepts = short_hypothesis(  
  brms_model, "Intercept = 0",  
  scope = "ranef", group = "L")  
  
random_slopes = short_hypothesis(  
  brms_model, "vt1 = 0",  
  scope = "ranef", group = "L")
```



Random Effects



```
> brms_model
Family: gaussian
Links: mu = identity; sigma = identity
Formula: height ~ vt1 + C + vt1:C + (vt1 + C + vt1:C | L) + (1 | A)
Data: exp_data (Number of observations: 2085)
Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 10
total post-warmup draws = 4000

Multilevel Hyperparameters:
~L (Number of levels: 15)

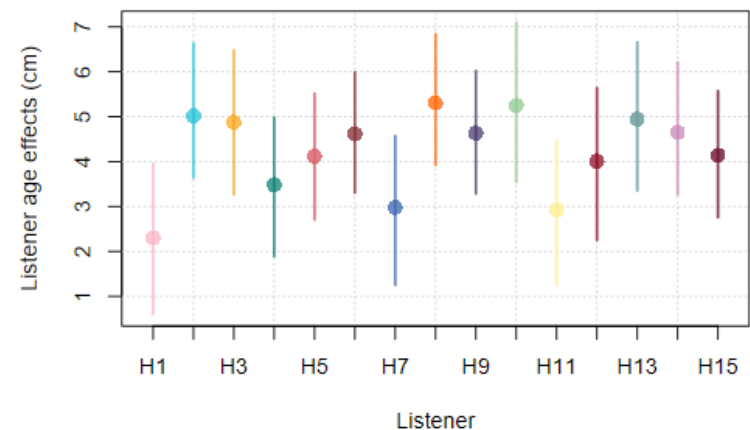
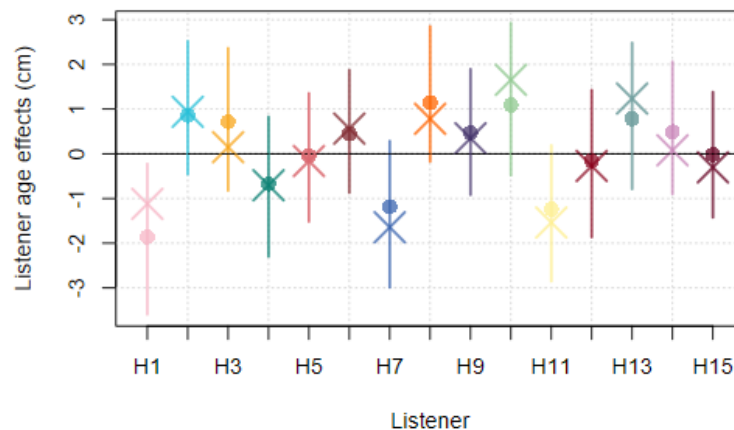
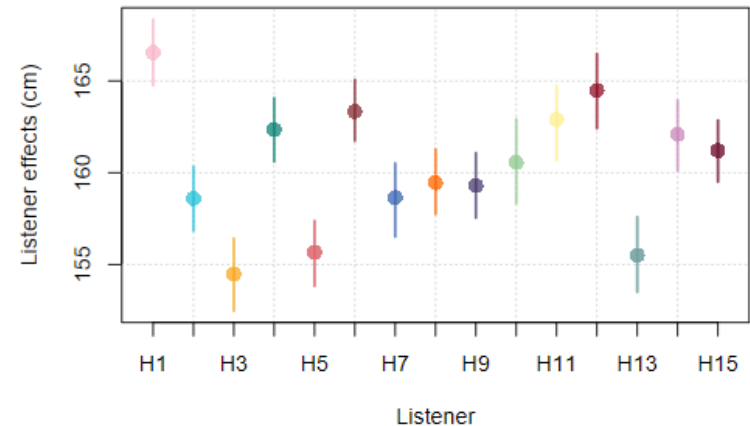
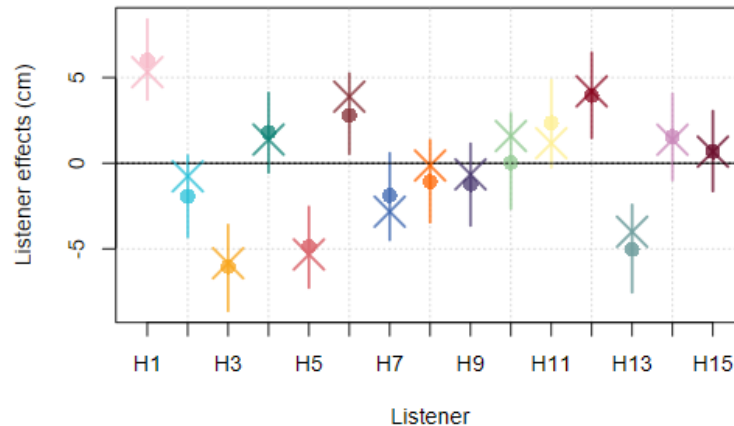
```

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat
sd(Intercept)	3.65	0.74	2.47	5.31	1.00
sd(vt1)	1.23	0.47	0.40	2.31	1.00
sd(C1)	4.51	0.91	3.02	6.54	1.00
sd(C2)	3.85	0.98	2.24	6.11	1.00
sd(C3)	3.37	0.90	1.85	5.37	1.00
sd(vt1:C1)	2.39	0.72	1.22	4.01	1.00
sd(vt1:C2)	1.81	0.78	0.37	3.49	1.00
sd(vt1:C3)	0.73	0.53	0.03	1.95	1.00
cor(Intercept, vt1)	-0.18	0.24	-0.62	0.29	1.00

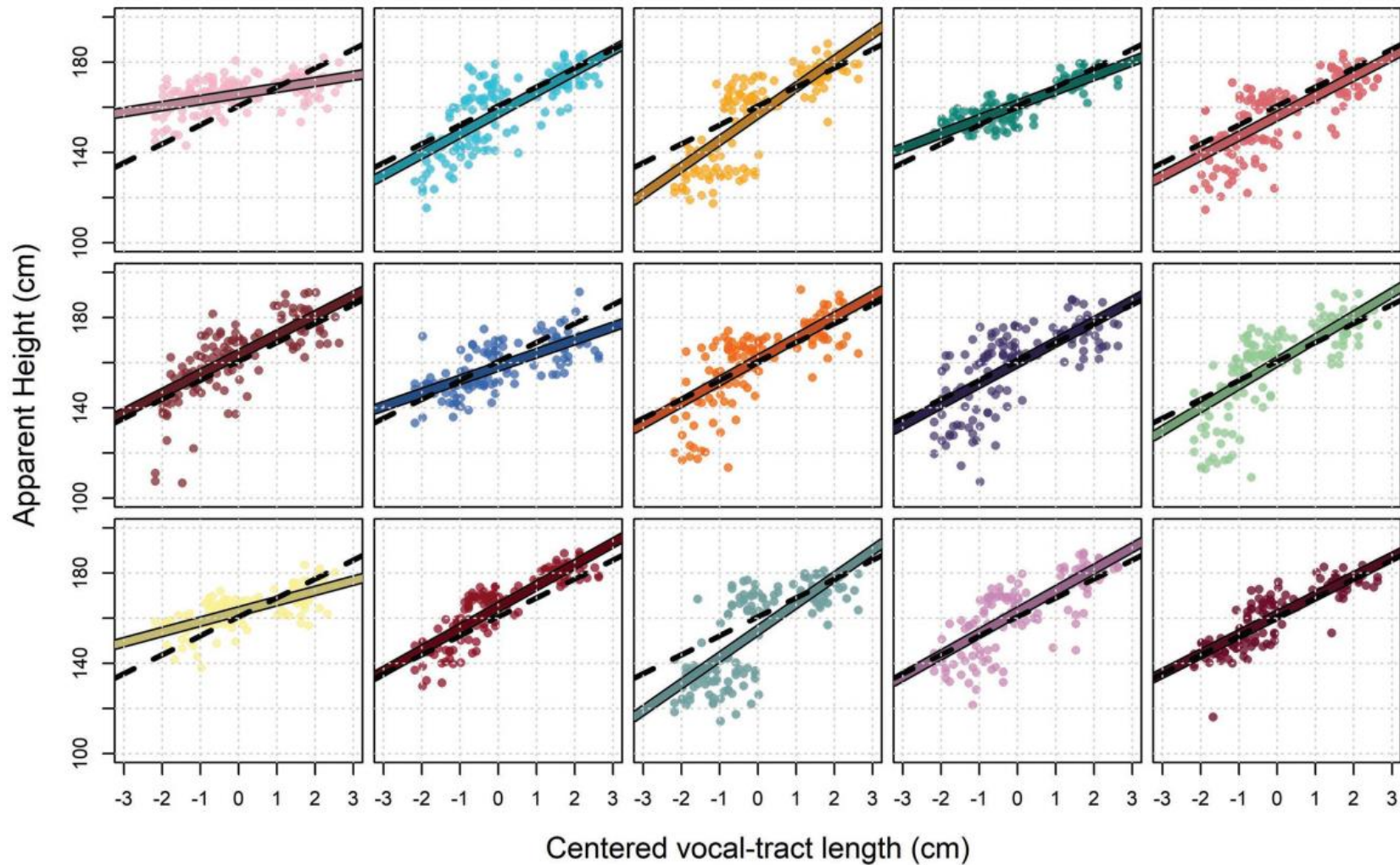
Random Effects (Ch. 5)

```
random_intercepts = short_hypothesis(  
  brms_model, "Intercept = 0",  
  scope = "ranef", group = "L")  
random_slopes = short_hypothesis(  
  brms_model, "vt1 = 0",  
  scope = "ranef", group = "L")
```

```
random_intercepts = short_hypothesis(  
  brms_model, "Intercept = 0",  
  scope = "coef", group = "L")  
random_slopes = short_hypothesis(  
  brms_model, "vt1 = 0",  
  scope = "coef", group = "L")
```



Listener-Specific Lines



Random Effects (Ch. 5)

- You can also get this using the ranef function.
- This returns a list of 3-dimensional tables of summarized random effects.

```
> random_effects = ranef(brms_model)$L
> random_effects
, , Intercept

      Estimate Est.Error      Q2.5      Q97.5
1    6.01076417  1.206563  3.7086748  8.3968151
2   -1.93798321  1.199241 -4.3108360  0.4523223
3   -6.05123381  1.275129 -8.6244935 -3.6056105
4    1.80652637  1.185608 -0.5485732  4.1059063
5   -4.87199708  1.220520 -7.2695741 -2.5483610
6    2.80344699  1.190432  0.5507715  5.2315839
7   -1.88324264  1.271838 -4.4830208  0.5837372
8   -1.08116867  1.214334 -3.4488263  1.3461112
9   -1.23743388  1.225204 -3.6410411  1.1363588
10   0.02878774  1.408835 -2.6557548  2.9141647
11   2.36009120  1.286413 -0.2449311  4.8587598
12   3.94868756  1.255440  1.4786338  6.4499939
13  -5.03057749  1.297564 -7.5382107 -2.4304268
14   1.55063967  1.264362 -0.9883924  4.0490636
15   0.67565279  1.172622 -1.6269754  3.0429613

, , vtl

      Estimate Est.Error      Q2.5      Q97.5
1  -1.86043563  0.8570942 -3.5924661 -0.2256687
2   0.85462941  0.7638298 -0.4583626  2.5166143
3   0.71496329  0.8151367 -0.8226958  2.3631150
4  -0.67543890  0.7759889 -2.2981784  0.8259511
5  -0.04372079  0.7165196 -1.5167818  1.3538206
6   0.45118599  0.6993539 -0.8712980  1.8731663
7  -1.18190919  0.8444629 -2.9883730  0.2853491
8   1.14315889  0.7617347 -0.1743833  2.8492609
```

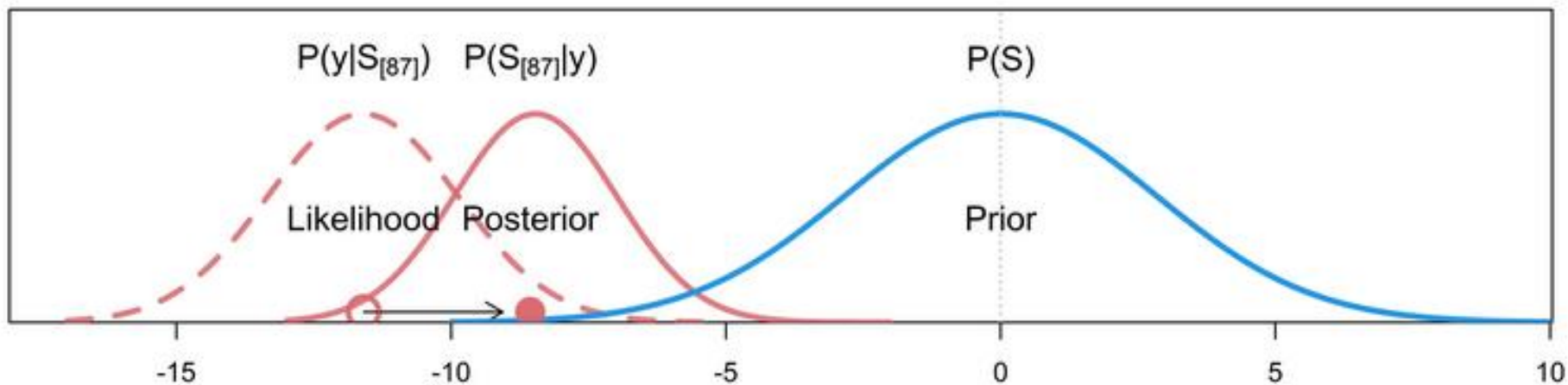
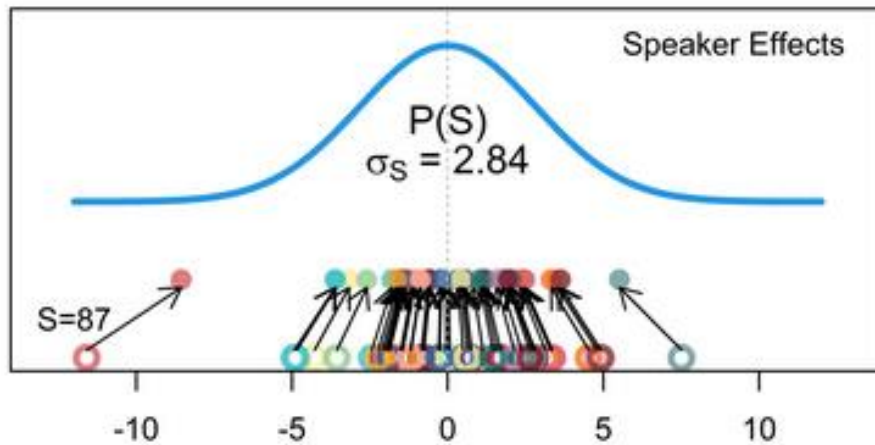
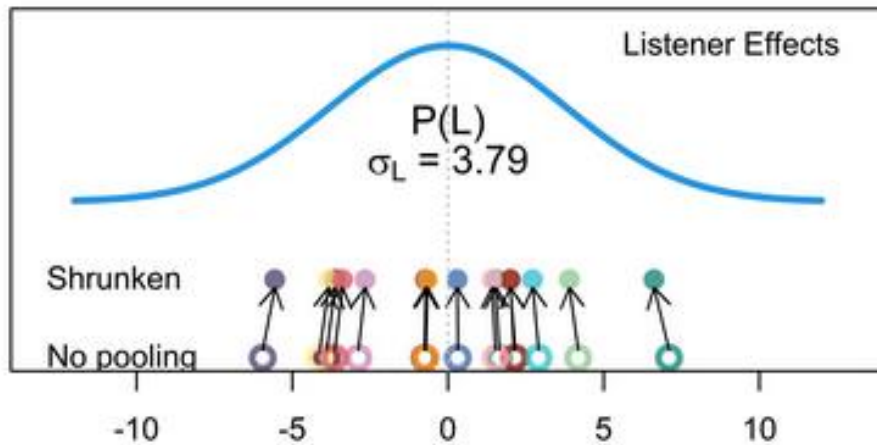
Random Effects (Ch. 5)

- If you set 'summary=FALSE', you get a list of 3-d matrices.
- Rows are individual samples, columns are effects.

```
> listener_effects = random_effects$L
> listener_effects
, , Intercept
```

	1	2	3	4	5	6	7
[1,]	4.154948	-2.8184474994	-7.661319	1.923444090	-4.4215853	3.3426075863	-2.339057404
[2,]	7.245795	-1.0693758870	-7.304954	1.940217937	-7.2694994	2.1905304593	-1.769245659
[3,]	4.587783	-3.7234058919	-6.829485	0.336166719	-4.6968929	1.6146247516	-1.332178138
[4,]	3.994830	-4.6803996053	-7.755466	0.292756760	-7.0304044	0.4090905927	-3.498596200
[5,]	3.665205	-5.4155052034	-7.782581	0.826631202	-6.5666178	1.3648355186	-3.763962459
[6,]	4.505377	-4.1082831006	-6.848761	2.410123883	-6.4049766	2.3656839027	-3.635893977
[7,]	4.499347	-2.9357655558	-6.611315	0.203715830	-5.7059018	1.2653714322	-2.455586715
[8,]	4.732438	-2.9792226934	-7.599259	-0.123840621	-5.9390719	1.1579052058	-4.581925738
[9,]	6.102063	-3.8468990499	-8.465508	0.864338830	-6.2922568	2.5679361258	-6.663665142
[10,]	5.897201	-2.0656644804	-7.684553	1.663973673	-5.9140121	3.4357181467	-1.479074288
[11,]	6.284814	-1.6361748039	-5.998162	1.356243835	-6.9250182	1.2812044393	-3.971978809
[12,]	6.573439	-1.4877599455	-5.614412	3.376106918	-3.6760293	3.6565980715	-0.456031032
[13,]	8.004622	-1.1604834944	-5.303222	2.875741897	-5.0025674	4.4705847459	-1.250115079
[14,]	6.315304	-1.5946635519	-6.493929	1.755900806	-4.3762488	3.0598228197	-0.202926163
[15,]	6.394682	-2.2296912499	-5.925962	1.839611873	-4.8066205	2.3506007171	-2.500985609
[16,]	5.407744	-2.7208090175	-8.829208	0.424960850	-4.8849291	1.9843116793	-3.166816023

Shrinkage (Ch. 5)



Exercises

1. Compare random effects for two listeners or speakers.
2. Manually combine random effects with a fixed effect to get the speaker or listener coefficient.

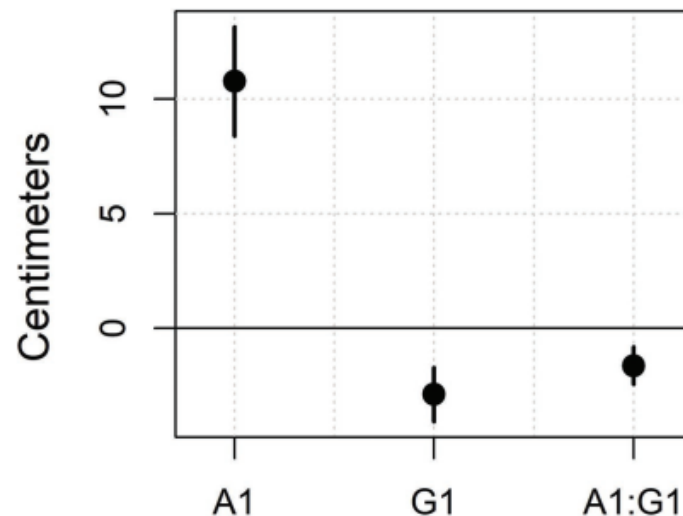
5. Reporting

Reporting Results

- For parameters or combinations of these, always report:
 - Means
 - Credible intervals (95% or other)
 - Standard deviations*

Reporting Results

Results indicate that the average apparent height across all speaker groups (i.e. the intercept) was 158.5 (s.d. = 1.12, 95% C.I. = [156.22, 160.62]). We also found an average effect of 10.8 cm for apparent speaker age (s.d. = 1.21, 95% C.I. = [8.39, 13.15]) and -2.9 cm for apparent speaker gender (s.d. = 0.6, 95% C.I. = [-4.07, -1.72]). In addition, we found an interaction between the effects of apparent age and apparent gender on apparent heights (mean = -1.64, s.d. = 0.41, 95% C.I. = [-2.44, -0.81]). The result of these effects is that apparent adults were perceived as taller than apparent children and apparent males were perceived as taller than apparent females. However, the difference in apparent height due to apparent gender was larger for adults than for children (and the effect for apparent age was larger for males than for females). Figure 7.11 presents the model's fixed effects other than the intercept (whose value is too large to plot in this range).



No such thing as “significance”

- Don't use the word 'significant'.
- 95% is meaningless.
- Things can be 'significant but' meaningless.
- Focus on interpretation, what likely matters, and how much you should believe it.

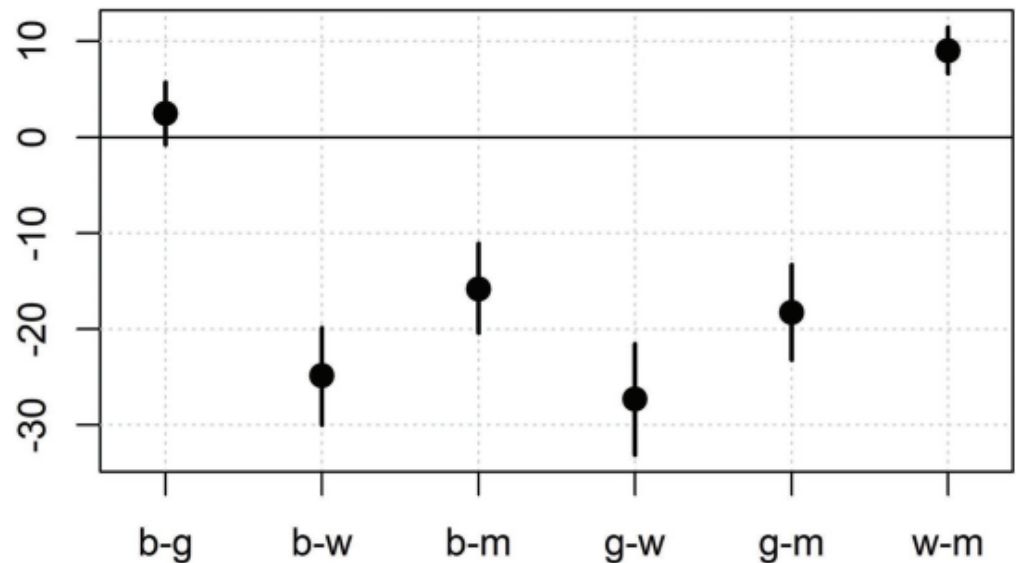
Pairwise Group Differences

```
# get group mean predictions
C = attributes(means_pred_interaction)$samples

# find pairwise differences
pairwise_diffs = cbind("b-g"=C[,1]-C[,2], "b-w"=C[,1]-C[,3],
                       "b-m"=C[,1]-C[,4], "g-w"=C[,2]-C[,3],
                       "g-m"=C[,2]-C[,4], "w-m"=C[,3]-C[,4])

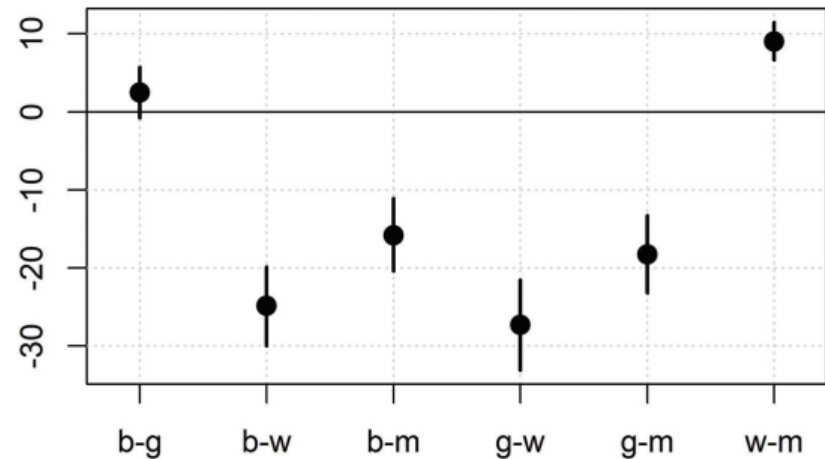
# summarize these
pairwise_diffs_summary = posterior_summary(pairwise_diffs)
```

Which are
'real'?



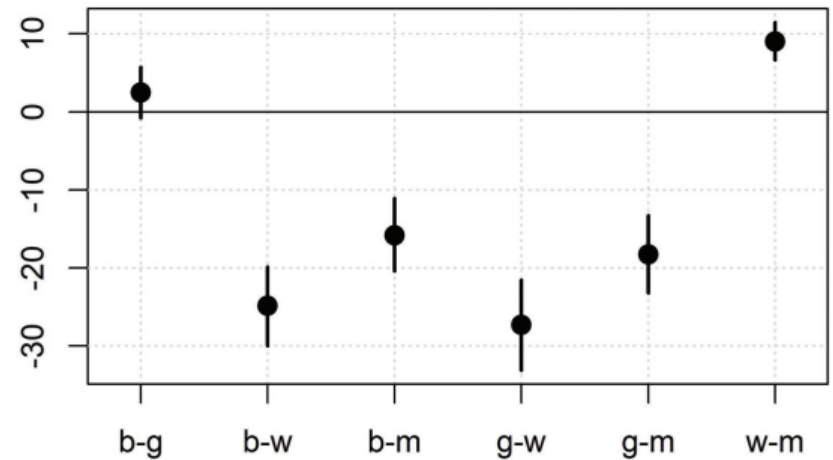
Is an Effect 'Real'? (Ch. 7)

- Most 95% Credible intervals don't cross 0. Does this make the effects 'real'?
- The b-g credible intervals crosses 0. Does this mean the effect equals 0?
- Even if an interval does cross zero, its most likely value is still not usually zero.



Type S and M Errors (Ch. 7)

- Better way to think of it:
Avoid type S and M errors.
- Type S (sign) errors: Is the value negative when you think its positive (or vice versa)?
- Type M (magnitude) errors: Is the value small when you think its large (or vice versa)?

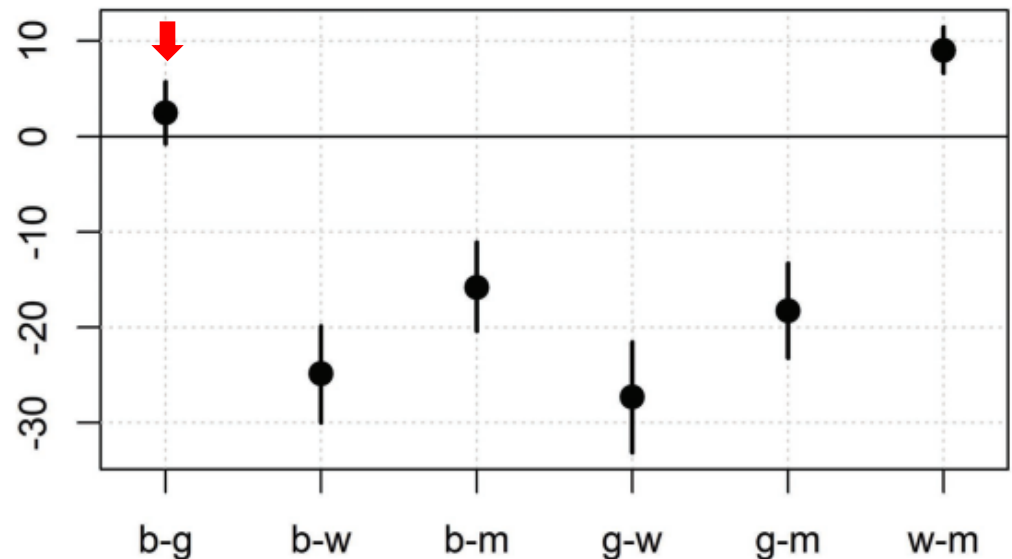


Region of Practical Equivalence (Ch. 7)

- Region of Practical Equivalence (ROPE): How small does an effect have to be before an effect *might as well* be zero?

```
# summary of difference between boy and girl means  
pairwise_diffs_summary[1,]  
##   Estimate Est.Error      Q2.5      Q97.5  
##    2.4643    1.6384   -0.7837    5.7199
```

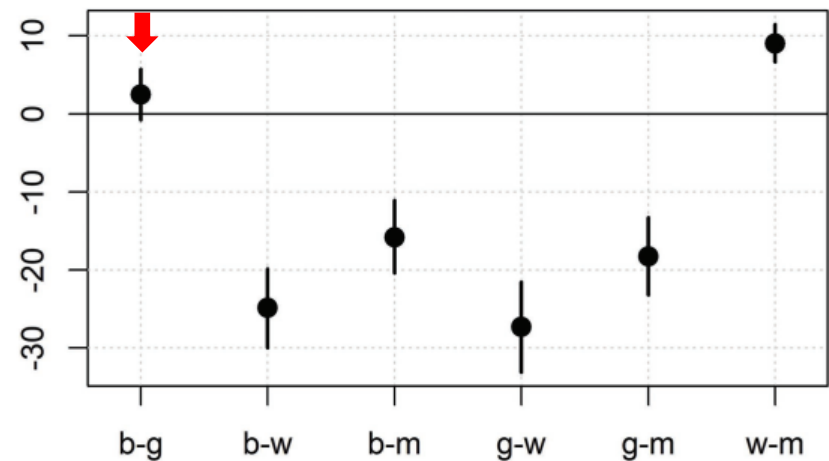
- Example: For human height, differences under 0.5 cm have little practical meaning. A 'significant' difference of 0.01 cm might as well be zero.



So, is the b-g Difference 'Real'? (Ch. 7)

- This is an epistemological or ontological claim.
- Statistics can help you make these claims, but you are responsible for them.
- Final word: The difference may be real and positive, but it is likely small. Further investigation is necessary.

```
# summary of difference between boy and girl means  
pairwise_diffs_summary[1,]  
##      Estimate Est.Error      Q2.5      Q97.5  
##      2.4643    1.6384    -0.7837    5.7199
```



Exercises

Present and interpret some results, parameters, or parameter comparisons.

Thank You!!